SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

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CSA1509 - CLOUD COMPUTING

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ASSESSMENT TOOL - 2

1) CLOUD Architecture :-

→ use cloud virtual Machines (VMs) to host the video streaming Application.

→ store videos in cloud object storage.

→ use content Delivery Network (CDN) to cache videos closer to users.

→ Deploy a load Balancer to distribute user Requests.

Load Balancing

→ Distributes incoming traffic across multiple servers.

→ Prevents server overload.

→ improves application availability and Response time.

conclusion :- A scalable cloud Architecture with CDN, load Balancing, and auto scaling provides smooth video streaming; better perform; and lower operational costs.

2) Solution :-

→ Package applications using Docker containers.

→ Deploy and manage containers using Kubernetes.

→ use multiple cloud Providers for better availability.

→ Kubernetes automatically identifies and connect Application.

Benefits :-

→ High Availablity.

→ vendor independence.

→ Better disaster Recovery.

→ Easy deployment and management.

Risks :-

→ Increased management complexity.

→ security challenges.

→ Higher Networking costs.

conclusion :-

container orchestration and with Kubernetes and multi-cloud deployment improves Application Relability, Scalablity and flexiblity

3) Backup strategy :-

→ Schedule automatic cloud backups.

→ Store multiple copies using Replication.

→ Enable versioning to Recover deleted file.

Disaster Recovery :-

→ Maintain backup copies in different cloud Regions.

→ Restore services quickly after failure.

→ Measures how quickly services must be Restored.

→ Achieved using automated failover and backup Recovery.

RPO (Recovery Point objective) :-

→ Measures Acceptable data loss.

→ Achieved through frequent backups and Replication.

conclusion :- cloud - native backup and disaster Recovery Ensure data Protection ; fast Recovery ; and uninterrupted business operations.

4) Hybrid cloud Design :-

→ store sensitive government data on - Premises.

→ Deploy non - sensitive applications in the Public cloud.

→ connect both Environments securely using VPN (or) dedicated links.

Identity Management :-

→ implement Identity and Access Management (IAM).

→ use Multi Factor Authentication (MFA).

→ Provide Role based Access control.

Encryption :-

→ Encrypt data during transmission.

→ Encrypt stored data using secure Encryption Methods.

→ Managing multiple Environments.

→ Ensuring Regulatory compliance.

→ Integrating between cloud and on - Premises systems.

conclusion :- Hybrid cloud provides security, flexibility and also compliance while allowing efficient use of cloud resources.

5) **Multi - Region Deployment :-**

→ Deploy the Application in multiple could Regions.

→ Distribute traffic using global load Balancer.

Data Replication :-

→ continuously Replicate data between Regions.

→ Keep all Regions synchronized.

Failover Mechanism :-

→ Automatically switch users to another Region if one Region fails.

→ Minimize downtime.

Benefits :-

→ High Availability.

→ Fault tolerance.

→ Improved Reliability.

→ Better user experience.

conclusion :- A multi factor Region deployment with Replication, failover, and continuous monitoring ensures High Availability & cloud services.